

Generation Interconnection System Impact Study Report

For Queue Project Q17-0852

Cornuta 100 kV
October 15, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0852. The Interconnection Customer proposes a 173.6 MW wind facility interconnecting at the Cornuta 100 kV substation in the Indiana area. The request was received on August 22, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 241 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$12,539,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Muldoon Mill / Hassen Cassel 100 kV L	Darlington Rd / Wayne Trace 100 kV	152.3	107	5.3
Hillcrest / Wayne Trace 100 kV	Hillcrest / Mudoon Mill 100 kV	127.7	180	5.1
Darlington Rd Substation / Hassen Cassel Substation 100 kV	Darlington Rd / Wayne Trace 100 kV	127.7	118	5.3
Monte Del Estado 6303 / Hrper 100 kV	Barret / Roseau County 100 kV Line	118.4	226	5.3
Cornuta / Industrial Park 100 kV	Dekalb Substation / Jones Chapel 500 kV	117.9	115	39.2
Columbia Hills / Indego 100 kV	Richland / Dekalb 500 kV L	114.4	148	6.0
Aep Pettit Avenue / Waynedale 100 kV Line 2	Muskogee / Montpelier Electric Generating Sta 500 kV	111.6	172	5.0
Lucent / 15Th Avenue South 230 kV	Omni-Hackberry / Norwich 500 kV	110.7	616	5.2
15Th Avenue South / Norwich 230 kV	Omni-Hackberry / Norwich 500 kV L	110.5	630	5.2
Richland 500 kV/230 kV Transformer	Troy Energy Llc / Richland 500 kV	108.0	2,150	12.6
Breed / Mcneil 230 kV L	Richland / Dekalb 500 kV	107.3	1,669	13.2
Sidney A Murray Jr Hydroelectric / Breed 230 kV	Dekalb / Richland 500 kV	107.3	1,669	13.2
Minnehaha / Sidney A Murray Jr Hydroelectric 230 kV	Richland / Dekalb 500 kV	107.2	1,138	8.3
Weedpatch / Canal Road 230 kV	Trego / Beaver 500 kV line	106.8	966	7.6
Rseau County / Barret 100 kV	Harper / Monte Del Estado 6303 100 kV	106.8	200	5.1
Slocum Rd / Mcneil 230 kV L	Richland / Dekalb 500 kV	106.3	1,662	13.2
Dekalb / Slocum Rd 230 kV	Richland Substation / Dekalb 500 kV	105.9	1,696	13.2

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Robison Park / Dekalb 230 kV	Jones Chapel / Dekalb 500 kV	105.1	2,339	24.1
Waynedale Substation / Aep Pettit Avenue 100 kV Line 1	Montpelier Electric Generating Sta / Muskogee 500 kV line	104.6	183	5.0
Highgate / Troy Energy Llc 230 kV L	Trego / Troy Energy Llc 500 kV	104.5	1,011	6.0
Post Rock / Newport Tie 230 kV	Beaer / Trego 500 kV line	104.3	1,691	12.9
Adams / Onancock Diesel Generation Facility 230 kV	Trego / Beaver Substation 500 kV L	103.7	2,118	14.2
Beaver / Trego 500 kV	Onancock Diesel Generation Facility / Adams 230 kV	103.7	4,242	29.1
Adams / Pickford 230 kV L	Trego / Beaver 500 kV	103.3	1,079	7.4
Allen / Aep Pettit Avenue 100 kV Line	Segs 5 / Hillrest 100 kV	103.2	298	8.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Muldoon Mill / Hassen Cassel 100 kV L	Rebuild line with higher-capacity conductor	\$3,335,000	27% (\$912,000)
Hillcrest / Wayne Trace 100 kV	Rebuild line with higher-capacity conductor	\$930,000	15% (\$138,000)
Hassen Cassel / Darlington Rd 100 kV	Rebuild line with higher-capacity conductor	\$3,807,000	27% (\$1,041,000)
Monte Del Estado 6303 / Hrper 100 kV	Rebuild line with higher-capacity conductor	\$5,863,000	8% (\$490,000)
Cornuta / Industrial Park 100 kV	Rebuild line with higher-capacity conductor	\$1,276,000	61% (\$775,000)
Columbia Hills / Indego 100 kV	Rebuild line with higher-capacity conductor	\$2,534,000	37% (\$939,000)
Waynedale / Aep Pettit Avenue 100 kV Ckt 2	Rebuild line with higher-capacity conductor	\$1,639,000	17% (\$276,000)
15Th Avenue South / Lucent Substation 230 kV line	Rebuild line with higher-capacity conductor	\$13,149,000	2% (\$308,000)
15Th Avenue S / Norwich 230 kV	Rebuild line with higher-capacity conductor	\$12,522,000	2% (\$293,000)
Richland 500-230 kV XFMR	Replace transformer with higher-rated unit	\$34,074,000	2% (\$546,000)
Mcneil / Breed 230 kV	Reconductor line	\$6,442,000	2% (\$131,000)
Sidney A Murray Jr Hydroelectric / Breed 230 kV	Reconductor line	\$15,377,000	2% (\$312,000)
Minnehaha / Sidney A Murray Jr Hydroelectric 230 kV	Reconductor line	\$7,982,000	2% (\$159,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Weedpatch / Cana Rd 230 kV	Reconductor line	\$6,234,000	2% (\$123,000)
Barret / Roseau County 100 kV	Reconductor line	\$2,175,000	10% (\$210,000)
Slocum Rd / Mcneil 230 kV L	Reconductor line	\$5,432,000	2% (\$110,000)
Dekalb / Slocum Rd 230 kV	Reconductor line	\$10,482,000	2% (\$213,000)
Robison Park / Dekalb 230 kV	Reconductor line	\$9,164,000	2% (\$224,000)
Waynedale Substation / Aep Pettit Avenue 100 kV Line 1	Reconductor line	\$641,000	17% (\$108,000)
Troy Energy Llc / Highgate Substation 230 kV	Reconductor line	\$17,125,000	2% (\$297,000)
Post Rock / Newport Tie 230 kV	Reconductor line	\$9,275,000	2% (\$181,000)
Adams / Onancock Diesel Generation Facility 230 kV	Reconductor line	\$24,991,000	2% (\$483,000)
Trego / Beaver 500 kV	Reconductor line	\$194,584,000	2% (\$3,710,000)
Adams / Pickford 230 kV L	Reconductor line	\$7,470,000	2% (\$146,000)
Allen / Aep Pettit Avenue 100 kV	Reconductor line	\$2,492,000	17% (\$414,000)
	Total allocated to project		\$12,539,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.